Technology Brief 5: Global Positioning System

The Global Positioning System (***GPS***), initially developed in the 1980s by the U.S. Department of Defense as a navigation tool for military use, has evolved into a system with numerous civilian applications, including vehicle tracking, aircraft navigation, map displays in automobiles and hand-held cell phones (**Fig. TF5-1**), and topographic mapping. The overall GPS comprises three segments. The ***space segment*** consists of 24 satellites (**Fig. TF5-2**), each circling Earth every 12 hours at an orbital altitude of about 12,000 miles and transmitting continuous coded time signals. All satellite transmitters broadcast coded messages at two specific frequencies: 1.57542 GHz and 1.22760 GHz. The ***user segment*** consists of hand-held or vehicle-mounted receivers that determine their own locations by receiving and processing multiple satellite signals. The third segment is a network of five ***ground stations*** distributed around the world that monitor the satellites and provide them with updates on their precise orbital information.

► GPS provides a location inaccuracy of about 30 m both horizontally and vertically, but it can be improved to within 1 m by ***differential GPS***. (See final section.) ◄



Figure TF5-1 iPhone map feature.

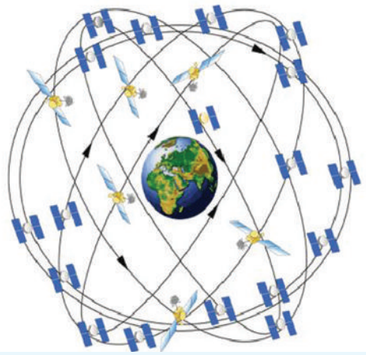


Figure TF5-2 GPS nominal satellite constellation. Four satellites in each plane with 20,200 km altitudes and a 55° inclination.

Principle of Operation

The ***triangulation technique*** allows the determination of the location (x_0, y_0, z_0) of any object in 3-D space from knowledge of the distances d_1, d_2, and d_3 between that object and three other independent points in space of known locations (x_1, y_1, z_1) to (x_3, y_3, z_3). In GPS, the distances are established by measuring the times it takes the signals to travel from the satellites to the GPS receivers, and then multiplying them by the speed of light $c = 3 \times 10^8$ m/s. Time synchronization is achieved by using atomic clocks. The satellites use very precise clocks, accurate to 3 nanoseconds (3×10^{-9} s), but receivers use less accurate, inexpensive, ordinary quartz clocks. Consequently, the receiver clock may have an unknown ***time offset error*** t_0 relative to the satellite clocks. *To correct for the time error of a GPS receiver, a signal from a fourth satellite is needed.*

技术简介 5：全球定位系统

全球定位系统（GPS）最初由美国国防部于 20 世纪 80 年代开发，作为军事用途的导航工具，现已发展成为具有多种民用应用的系统，包括车辆跟踪、飞机导航、汽车和手持手机中的地图显示（图 TF5-1）以及地形测绘。整个 GPS 包括三个部分。太空部分由 24 颗卫星组成（图 TF5-2），每颗卫星每 12 小时绕地球一周，轨道高度约为 12,000 英里，并传输连续编码时间信号。所有卫星发射机都以两个特定频率广播编码消息：1.57542 GHz 和 1.22760 GHz。用户部分由手持式或车载接收器组成，它们通过接收和处理多个卫星信号来确定自己的位置。第三部分是由分布在世界各地的五个地面站组成的网络，用于监视卫星并为其提供最新的精确轨道信息。

► GPS 的水平和垂直定位误差约为 30 m，但差分 GPS 可以将其改善到 1 m 以内。（参见最后一节。）◄



图TF5-1 iPhone地图功能。

工作原理

三角测量技术允许根据已知位置 $(x_1、y_1、z_1)$ 到 (x_3) 的空间中的其他三个独立点与该对象之间的距离 $d_1、d_2$ 和 d_3 的知识，确定 3D 空间中任何对象的位置 $(x_0、y_0、z_0)$，y_3，z_3)。在 GPS 中，距离是通过测量信号从卫星传输到 GPS 接收器所需的时间，然后乘以光速 $c = 3 \times 10^8$ m/s 来确定的。时间同步是通过使用原子钟来实现的。卫星使用非常精确的时钟，精确到 3 纳秒（3×10^{-9} s），但接收器使用不太精确、廉价的普通石英钟。因此，接收器时钟相对于卫星时钟可能具有未知的时间偏移误差 t_0。为了纠正 GPS 接收器的时间误差，需要来自第四颗卫星的信号。

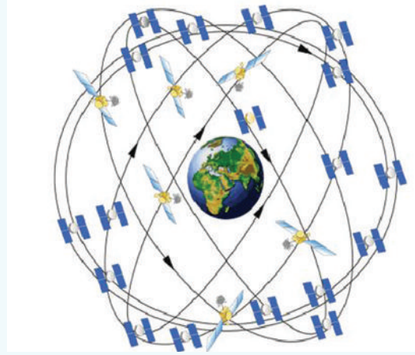


图 TF5-2 GPS 标称卫星星座。每个平面有四颗卫星，高度为 20,200 公里，倾角为 55°。

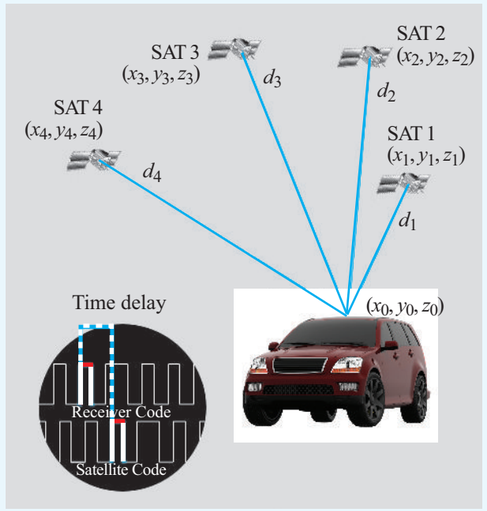


Figure TF5-3 Automobile GPS receiver at location (x_0, y_0, z_0).

The GPS receiver of the automobile in **Fig. TF5-3** is at distances d_1 to d_4 from the GPS satellites. Each satellite sends a message identifying its orbital coordinates (x_1, y_1, z_1) for satellite 1, and so on for the other satellites, together with a binary-coded sequence common to all satellites. The GPS receiver generates the same binary sequence (**Fig. TF5-3**), and by comparing its code with the one received from satellite 1, it determines the time t_1 corresponding to travel time over the distance d_1. A similar process applies to satellites 2 to 4, leading to four equations:

$$d_1^2 = (x_1 - x_0)^2 + (y_1 - y_0)^2 + (z_1 - z_0)^2 = c^2\,[(t_1 + t_0)]^2,$$
$$d_2^2 = (x_2 - x_0)^2 + (y_2 - y_0)^2 + (z_2 - z_0)^2 = c^2\,[(t_2 + t_0)]^2,$$
$$d_3^2 = (x_3 - x_0)^2 + (y_3 - y_0)^2 + (z_3 - z_0)^2 = c^2\,[(t_3 + t_0)]^2,$$
$$d_4^2 = (x_4 - x_0)^2 + (y_4 - y_0)^2 + (z_4 - z_0)^2 = c^2\,[(t_4 + t_0)]^2.$$

The four satellites report their coordinates (x_1, y_1, z_1) to (x_4, y_4, z_4) to the GPS receiver, and the time delays t_1 to t_4 are measured directly by the receiver. The unknowns are (x_0, y_0, z_0), the coordinates of the GPS receiver, and the time offset of its clock t_0. Simultaneous solution of the four equations provides the desired location information.

Differential GPS

The 30 m GPS position inaccuracy is attributed to several factors, including ***time-delay errors*** (due to the difference between the speed of light and the actual signal speed in the troposphere) that depend on the receiver's location on Earth, delays due to signal reflections by tall buildings, and satellites' locations misreporting errors.

► ***Differential GPS***, or ***DGPS***, uses a stationary reference receiver at a location with known coordinates. ◄

By calculating the difference between its location on the basis of the GPS estimate and its true location, the ***reference receiver*** establishes coordinate correction factors and transmits them to all DGPS receivers in the area. Application of the correction information usually reduces the location inaccuracy down to about 1 m.

152 CHAPTER 3 VECTOR ANALYSIS

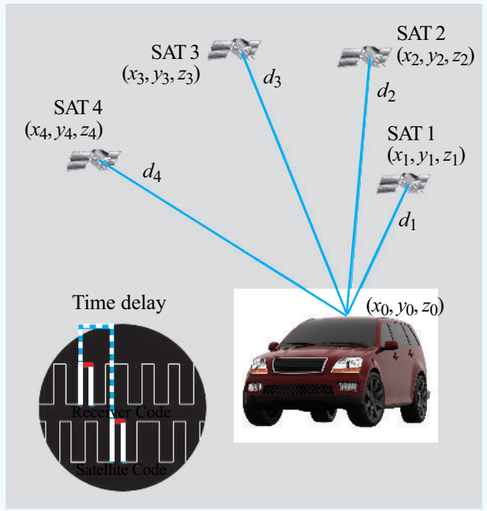


图 TF5-3 位置 (x_0,y_0,z_0) 处的汽车 GPS 接收器。

图TF5-3中汽车的GPS接收器距GPS卫星的距离为d_1至d_4。每颗卫星发送一条消息，标识卫星 1 的轨道坐标（x_1、y_1、z_1），其他卫星依此类推，以及所有卫星共用的二进制编码序列。 GPS接收器生成相同的二进制序列（图TF5-3），并通过将其代码与从卫星1接收到的代码进行比较，确定与经过距离d_1的行进时间相对应的时间t_1。一个

类似的过程适用于卫星 2 至 4，从而得出四个方程：

$$d_1^2 = (x_1 - x_0)^2 + (y_1 - y_0)^2 + (z_1 - z_0)^2 = c^2\,[(t_1 + t_0)]^2,$$
$$d_2^2 = (x_2 - x_0)^2 + (y_2 - y_0)^2 + (z_2 - z_0)^2 = c^2\,[(t_2 + t_0)]^2,$$
$$d_3^2 = (x_3 - x_0)^2 + (y_3 - y_0)^2 + (z_3 - z_0)^2 = c^2\,[(t_3 + t_0)]^2,$$
$$d_4^2 = (x_4 - x_0)^2 + (y_4 - y_0)^2 + (z_4 - z_0)^2 = c^2\,[(t_4 + t_0)]^2.$$

四颗卫星将它们的坐标（x_1、y_1、z_1）到（x_4、y_4、z_4）报告给 GPS 接收器，并且时间延迟 t_1 到 t_4 由接收器直接测量。未知数为 (x_0, y_0, z_0)、GPS 接收器的坐标及其时钟的时间偏移 t_0。四个方程的同时求解提供了所需的位置信息。

差分GPS

30 m GPS 位置误差归因于多种因素，包括取决于接收器在地球上的位置的时延误差（由于光速与对流层中实际信号速度之间的差异）、高层建筑反射信号造成的延迟以及卫星位置误报误差。

► 差分 GPS 或 DGPS 在已知坐标的位置使用固定参考接收器。 ◄

通过根据 GPS 估计计算其位置与其真实位置之间的差异，参考接收器建立坐标校正因子并将其传输到该区域中的所有 DGPS 接收器。校正信息的应用通常可将定位误差降低至约 1 m。

Exercise 3-11: The power density radiated by a star (**Fig. E3.11(a)**) decreases radially as $S(R) = S_0/R^2$, where R is the radial distance from the star and S_0 is a constant. Recalling that the gradient of a scalar function denotes the maximum rate of change of that function per unit distance and the direction of the gradient is along the direction of maximum increase, generate an arrow representation of ∇S.

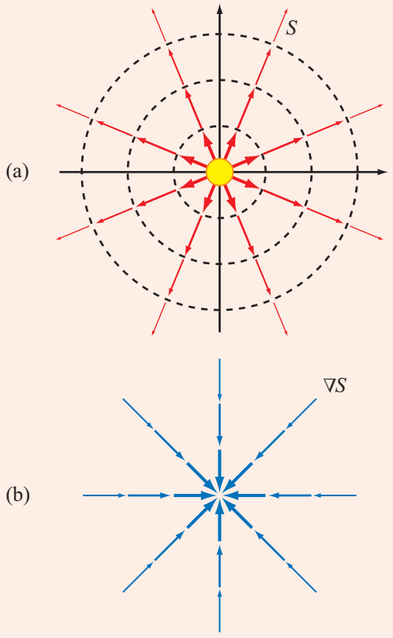


Figure E3.11

Answer: $\nabla S = -\hat{\mathbf{R}}\, 2S_0/R^3$ (**Fig. E3.11(b)**). (See EM.)

Exercise 3-12: The graph in **Fig. E3.12(a)** depicts a gentle change in atmospheric temperature from T_1 over the sea to T_2 over land. The temperature profile is described by the function

$$T(x) = T_1 + \frac{T_2 - T_1}{e^{-x} + 1},$$

where x is measured in kilometers and $x = 0$ is the sea–land boundary. (a) In which direction does ∇T point and (b) at what value of x is it a maximum?

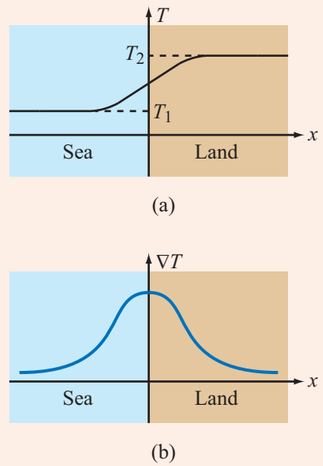


Figure E3.12

Answer: (a) $+\hat{\mathbf{x}}$; (b) at $x = 0$.

$$T(x) = T_1 + \frac{T_2 - T_1}{e^{-x} + 1},$$

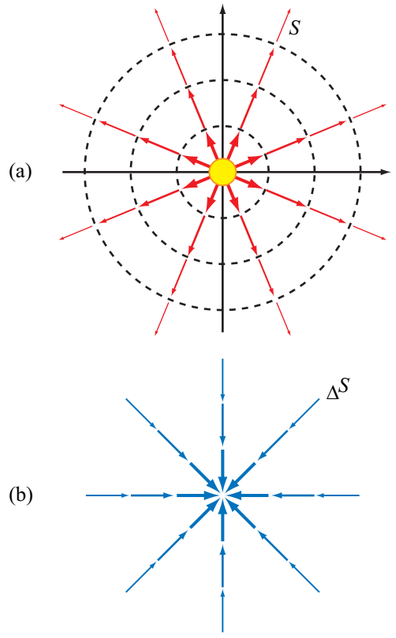
$$\nabla T = \hat{\mathbf{x}}\,\frac{\partial T}{\partial x} = \hat{\mathbf{x}}\,\frac{e^{-x}(T_2 - T_1)}{(e^{-x} + 1)^2}.$$

(See EM.)

3-5 Divergence of a Vector Field

From our brief introduction of Coulomb's law in Chapter 1, we know that an isolated, positive point charge q induces an electric field $\mathbf{E}$ in the space around it with the direction of $\mathbf{E}$ being outward away from the charge. Also, the strength (magnitude) of $\mathbf{E}$ is proportional to q and decreases with distance R from the charge as $1/R^2$. In a graphical presentation, a vector field is usually represented by ***field lines***, as shown in **Fig. 3-21**. The arrowhead denotes the direction of the field at the point where the field line is drawn, and the length of the line provides a qualitative depiction of the field's magnitude.

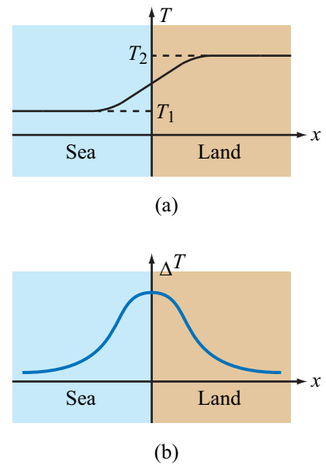
练习 3-11：恒星辐射的功率密度（图 E3.11(a)）随着 $S(R) = S_0/R^2$ 径向减小，其中 R 是距恒星的径向距离，S_0 是常数。回想一下标量函数的梯度表示该函数每单位距离的最大变化率，并且梯度的方向沿着最大增加的方向，生成 ∇S 的箭头表示。



图E3.11

答案：$\nabla S = -\hat{\mathrm{R}}\, 2S_0/R^3$ (图E3.11(b))。 （参见EM。）

其中 x 以公里为单位，$x = 0$ 是海陆边界。 (a) ∇T 指向哪个方向，以及 (b) x 的最大值在多少处？



图E3.12

答案：(a) $+\hat{\mathrm{x}}$； (b) 在 $x = 0$ 处。

$$T(x) = T_1 + \frac{T_2 - T_1}{e^{-x} + 1},$$

$$\nabla T = \hat{\mathbf{x}}\,\frac{\partial T}{\partial x} = \hat{\mathbf{x}}\,\frac{e^{-x}(T_2 - T_1)}{(e^{-x} + 1)^2}.$$

(参见EM。)

练习3-12：图E3.12(a)中的图表描绘了大气温度从海上的T_1到陆地上的T_2的缓慢变化。温度曲线由函数描述

$$T(x) = T_1 + \frac{T_2 - T_1}{e^{-x} + 1},$$

3-5 矢量场的散度

从第一章对库仑定律的简要介绍中，我们知道一个孤立的正点电荷 q 在其周围的空间中感应出一个电场 E，E 的方向远离电荷。此外，E 的强度（大小）与 q 成正比，并随着距电荷的距离 R 的增加而减小，即 $1/R^2$。在图形表示中，矢量场通常用场线表示，如图 3-21 所示。箭头表示绘制场线处的场方向，线的长度提供了场大小的定性描述。

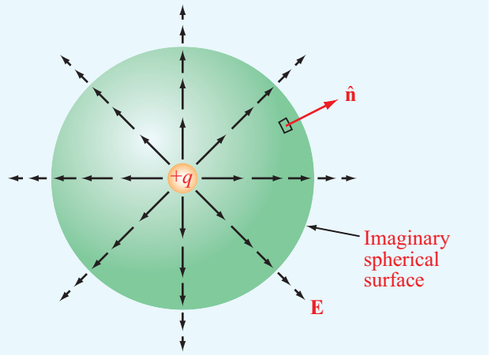


Figure 3-21 Flux lines of the electric field **E** due to a positive charge q.

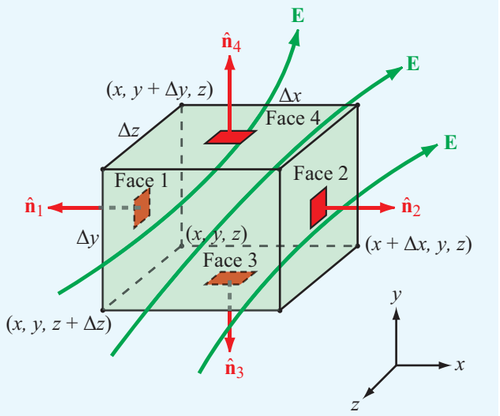


Figure 3-22 Flux lines of a vector field **E** passing through a differential rectangular parallelepiped of volume $\Delta\mathcal{U} = \Delta x\,\Delta y\,\Delta z$.

At a surface boundary, ***flux density*** is defined as the amount of outward flux crossing a unit surface ds:

$$\text{Flux density of } \mathbf{E} = \frac{\mathbf{E}\cdot d\mathbf{s}}{|d\mathbf{s}|} = \frac{\mathbf{E}\cdot\hat{\mathbf{n}}\,ds}{ds} = \mathbf{E}\cdot\hat{\mathbf{n}}, \tag{3.85}$$

where $\hat{\mathbf{n}}$ is the normal to $d\mathbf{s}$. The ***total flux*** outwardly crossing a closed surface S, such as the enclosed surface of the imaginary sphere outlined in **Fig. 3-21**, is

$$\text{Total flux} = \oint_S \mathbf{E}\cdot d\mathbf{s}. \tag{3.86}$$

Let us now consider the case of a differential rectangular parallelepiped, such as a cube, whose edges align with the Cartesian axes shown in **Fig. 3-22**. The edges are of lengths Δx along x, Δy along y, and Δz along z. A vector field $\mathbf{E}(x, y, z)$ exists in the region of space containing the parallelepiped, and we wish to determine the flux of **E** through its total surface S. Since S includes six faces, we need to sum up the fluxes through all of them, and by definition, the flux through any face is the *outward* flux from the volume $\Delta\mathcal{U}$ through that face.

Let **E** be defined as

$$\mathbf{E} = \hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z. \tag{3.87}$$

The area of the face marked 1 in **Fig. 3-22** is $\Delta y\,\Delta z$, and its unit vector $\hat{\mathbf{n}}_1 = -\hat{\mathbf{x}}$. Hence, the outward flux F_1 through face 1 is

$$\begin{aligned} F_1 &= \int_{\text{Face 1}} \mathbf{E}\cdot\hat{\mathbf{n}}_1\,ds \\ &= \int_{\text{Face 1}} (\hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z)\cdot(-\hat{\mathbf{x}})\,dy\,dz \\ &\approx -E_x(1)\,\Delta y\,\Delta z, \end{aligned} \tag{3.88}$$

where $E_x(1)$ is the value of E_x at the center of face 1. Approximating E_x over face 1 by its value at the center is justified by the assumption that the differential volume under consideration is very small.

Similarly, the flux out of face 2 (with $\hat{\mathbf{n}}_2 = \hat{\mathbf{x}}$) is

$$F_2 = E_x(2)\,\Delta y\,\Delta z, \tag{3.89}$$

where $E_x(2)$ is the value of E_x at the center of face 2. Over a differential separation Δx between the centers of faces 1 and 2, $E_x(2)$ is related to $E_x(1)$ by

$$E_x(2) = E_x(1) + \frac{\partial E_x}{\partial x}\,\Delta x, \tag{3.90}$$

where we have ignored higher-order terms involving $(\Delta x)^2$ and higher powers because their contributions are negligibly small when Δx is very small. Substituting Eq. (3.90) into Eq. (3.89) gives

$$F_2 = \left[E_x(1) + \frac{\partial E_x}{\partial x}\,\Delta x\right]\,\Delta y\,\Delta z. \tag{3.91}$$

The sum of the fluxes out of faces 1 and 2 is obtained by adding Eqs. (3.88) and (3.91),

$$F_1 + F_2 = \frac{\partial E_x}{\partial x}\,\Delta x\,\Delta y\,\Delta z. \tag{3.92a}$$

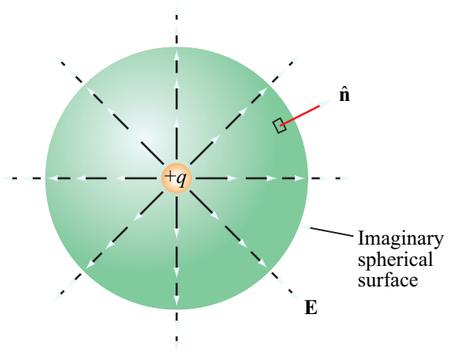


图 3-21 正电荷 q 产生的电场 E 的磁通线。

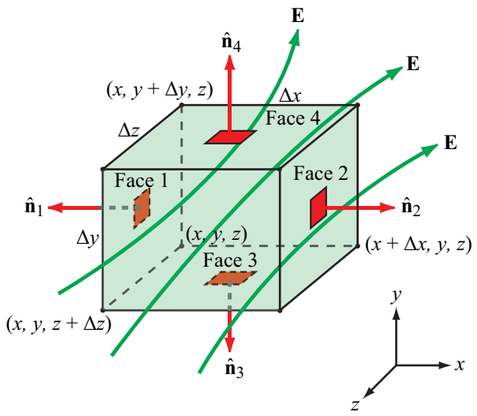


图 3-22 穿过体积为 $\Delta\mathcal{U} = \Delta x\,\Delta y\,\Delta z$ 的微分长方体的矢量场 E 的磁通线。

在表面边界处，通量密度定义为穿过单位表面 ds 的向外通量：

$$\text{Flux density of } \mathbf{E} = \frac{\mathbf{E}\cdot d\mathbf{s}}{|d\mathbf{s}|} = \frac{\mathbf{E}\cdot\hat{\mathbf{n}}\,ds}{ds} = \mathbf{E}\cdot\hat{\mathbf{n}}, \tag{3.85}$$

其中 n̂ 是 ds 的法线。向外穿过封闭表面 S（如图 3-21 所示的假想球体的封闭表面）的总通量为

$$\text{Total flux} = \oint_S \mathbf{E}\cdot d\mathbf{s}. \tag{3.86}$$

现在让我们考虑微分长方体的情况，例如立方体，其边缘与笛卡尔轴对齐，如图 3-22 所示。边沿 x 的长度为 Δx，沿 y 的长度为 Δy，沿 z 的长度为 Δz。向量场 E(x, y, z) 存在于包含平行六面体的空间区域中，我们希望确定 E 通过其总表面 S 的通量。由于 S 包括六个面，我们需要对通过所有面的通量求和，根据定义，通过任何面的通量是从体积 Δυ 通过该面的向外通量。

令 E 定义为

$$\mathbf{E} = \hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z. \tag{3.87}$$

图3-22中标记为1的面的面积为ΔyΔz，其单位向量为n̂ $_1 = -\hat{\mathrm{x}}$。因此，通过面 1 的向外通量 F_1 为

$$\begin{aligned} F_1 &= \int_{\text{Face 1}} \mathbf{E}\cdot\hat{\mathbf{n}}_1\,ds \\ &= \int_{\text{Face 1}} (\hat{\mathbf{x}}E_x + \hat{\mathbf{y}}E_y + \hat{\mathbf{z}}E_z)\cdot(-\hat{\mathbf{x}})\,dy\,dz \\ &\approx -E_x(1)\,\Delta y\,\Delta z, \end{aligned} \tag{3.88}$$

其中 $E_x(1)$ 是面 1 中心处的 E_x 值。通过假设所考虑的微分体积非常小，可以通过面 1 中心处的值来近似 E_x。

类似地，面 2（具有 $\hat{\mathrm{n}}_2 = \hat{\mathrm{x}}$）的通量为

$$F_2 = E_x(2)\,\Delta y\,\Delta z, \tag{3.89}$$

其中 $E_x(2)$ 是面 2 中心处的 E_x 值。在面 1 和 2 中心之间的差分间隔 Δx 上，$E_x(2)$ 与 $E_x(1)$ 的关系为

$$E_x(2) = E_x(1) + \frac{\partial E_x}{\partial x}\,\Delta x, \tag{3.90}$$

我们忽略了涉及 $(\Delta x)^2$ 和更高幂的高阶项，因为当 Δx 非常小时，它们的贡献可以忽略不计。代入方程式（3.90）代入方程。 (3.89) 给出

$$F_2 = \left[E_x(1) + \frac{\partial E_x}{\partial x}\,\Delta x\right]\,\Delta y\,\Delta z. \tag{3.91}$$

面 1 和面 2 的通量之和可通过添加等式获得。 （3.88）和（3.91），

$$F_1 + F_2 = \frac{\partial E_x}{\partial x}\,\Delta x\,\Delta y\,\Delta z. \tag{3.92a}$$

Repeating the same procedure to each of the other face pairs leads to

$$F_3 + F_4 = \frac{\partial E_y}{\partial y}\,\Delta x\,\Delta y\,\Delta z, \tag{3.92b}$$

$$F_5 + F_6 = \frac{\partial E_z}{\partial z}\,\Delta x\,\Delta y\,\Delta z. \tag{3.92c}$$

The sum of fluxes F_1 through F_6 gives the total flux through surface S of the parallelepiped:

$$\oint_S \mathbf{E}\cdot d\mathbf{s} = \left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}\right)\Delta x\,\Delta y\,\Delta z = (\text{div }\mathbf{E})\,\Delta\upsilon, \tag{3.93}$$

where $\Delta\upsilon = \Delta x\,\Delta y\,\Delta z$ and div $\mathbf{E}$ is a scalar function called the ***divergence*** of $\mathbf{E}$, specified in Cartesian coordinates as

$$\text{div }\mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}. \tag{3.94}$$

► By shrinking the volume $\Delta\upsilon$ to zero, we define the ***divergence*** of $\mathbf{E}$ at a point as the net outward flux per unit volume over a closed incremental surface. ◄

Thus, from Eq. (3.93), we have

$$\text{div }\mathbf{E} \triangleq \lim_{\Delta\upsilon\to 0}\frac{\oint_S \mathbf{E}\cdot d\mathbf{s}}{\Delta\upsilon}, \tag{3.95}$$

where S encloses the elemental volume $\Delta\upsilon$. Instead of denoting the divergence of $\mathbf{E}$ by div $\mathbf{E}$, it is common practice to denote it as $\nabla\cdot\mathbf{E}$. That is,

$$\nabla\cdot\mathbf{E} = \text{div }\mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \tag{3.96}$$

for a vector $\mathbf{E}$ in Cartesian coordinates.

► From the definition of the divergence of $\mathbf{E}$ given by Eq. (3.95), field $\mathbf{E}$ has positive divergence if the net flux out of surface S is positive, which may be "viewed" as if volume $\Delta\upsilon$ contains a *source* of field lines. If the divergence is negative, $\Delta\upsilon$ may be viewed as containing a *sink* of field lines because the net flux is into $\Delta\upsilon$. For a uniform field $\mathbf{E}$, the same amount of flux enters $\Delta\upsilon$ as leaves it; hence, its divergence is zero and the field is said to be ***divergenceless***. ◄

The divergence is a differential operator, it always operates on vectors, and the result of its operation is a scalar. This is in contrast with the gradient operator, which always operates on scalars and results in a vector. Expressions for the divergence of a vector in cylindrical and spherical coordinates are provided in Appendix C.

The divergence operator is distributive. That is, for any pair of vectors $\mathbf{E}_1$ and $\mathbf{E}_2$,

$$\nabla\cdot(\mathbf{E}_1 + \mathbf{E}_2) = \nabla\cdot\mathbf{E}_1 + \nabla\cdot\mathbf{E}_2. \tag{3.97}$$

If $\nabla\cdot\mathbf{E} = 0$, the vector field $\mathbf{E}$ is called ***divergenceless***.

The result given by Eq. (3.93) for a differential volume $\Delta\upsilon$ can be extended to relate the volume integral of $\nabla\cdot\mathbf{E}$ over any volume υ to the flux of $\mathbf{E}$ through the closed surface S that bounds υ. That is,

$$\int_\upsilon \nabla\cdot\mathbf{E}\,d\upsilon = \oint_S \mathbf{E}\cdot d\mathbf{s}. \tag{3.98}$$

(divergence theorem)

This relationship, known as the ***divergence theorem***, is used extensively in electromagnetics.

Example 3-13: Calculating the Divergence

Determine the divergence of each of the following vector fields and then evaluate them at the indicated points:

(a) $\mathbf{E} = \hat{\mathbf{x}}3x^2 + \hat{\mathbf{y}}2z + \hat{\mathbf{z}}x^2z$ at $(2,-2,0)$;
(b) $\mathbf{E} = \hat{\mathbf{R}}(a^3\cos\theta/R^2) - \hat{\boldsymbol{\theta}}(a^3\sin\theta/R^2)$ at $(a/2, 0, \pi)$.

Solution: (a)

$$\begin{aligned}\nabla\cdot\mathbf{E} &= \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}\\ &= \frac{\partial}{\partial x}(3x^2) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}(x^2z)\\ &= 6x + 0 + x^2 = x^2 + 6x.\end{aligned}$$

At $(2,-2,0)$, $\nabla\cdot\mathbf{E}\big|_{(2,-2,0)} = 16.$

对其他每对面孔重复相同的过程会导致

$$F_3 + F_4 = \frac{\partial E_y}{\partial y}\,\Delta x\,\Delta y\,\Delta z, \tag{3.92b}$$

$$F_5 + F_6 = \frac{\partial E_z}{\partial z}\,\Delta x\,\Delta y\,\Delta z. \tag{3.92c}$$

通量 F_1 到 F_6 的总和给出了通过平行六面体表面 S 的总通量:

$$\oint_S \mathbf{E}\cdot d\mathbf{s} = \left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}\right)\Delta x\,\Delta y\,\Delta z = (\text{div }\mathbf{E})\,\Delta\upsilon, \tag{3.93}$$

其中 $\Delta\upsilon = \Delta x\,\Delta y\,\Delta z$ 和 div E 是称为 E 散度的标量函数，在笛卡尔坐标中指定为

$$\text{div }\mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}. \tag{3.94}$$

► 通过将体积 $\Delta\upsilon$ 缩小到零，我们将 E 在某一点的散度定义为闭合增量表面上每单位体积的净向外通量。◄

因此，从方程。 (3.93) ，我们有

$$\text{div }\mathbf{E} \triangleq \lim_{\Delta\upsilon\to 0}\frac{\oint_S \mathbf{E}\cdot d\mathbf{s}}{\Delta\upsilon}, \tag{3.95}$$

其中 S 包含元素体积 $\Delta\upsilon$ 。通常的做法不是用 div E 来表示 E 的散度，而是将其表示为 $\nabla\cdot$ E。也就是说，

$$\nabla\cdot\mathbf{E} = \text{div }\mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \tag{3.96}$$

对于笛卡尔坐标系中的向量 E。

► 根据方程给出的 E 散度的定义： (3.95) ，如果离开表面 S 的净通量为正，则场 E 具有正散度，这可以被“视为”体积 $\Delta\upsilon$ 包含场线源。如果散度为负，则 $\Delta\upsilon$ 可被视为包含磁力线汇，因为净通量位于 $\Delta\upsilon$ 中。对于均匀场 E，进入 $\Delta\upsilon$ 的通量与离开它的通量相同；因此，它的散度为零，并且该场被称为无散度。 ◄

散度是一个微分算子，它总是对向量进行运算，其运算结果是一个标量。这与梯度运算符相反，梯度运算符始终对标量进行运算并产生向量。附录 C 提供了柱坐标和球坐标中向量散度的表达式。

散度算子是分布的。也就是说，对于任意一对向量 E_1 和 E_2，

$$\nabla\cdot(\mathbf{E}_1 + \mathbf{E}_2) = \nabla\cdot\mathbf{E}_1 + \nabla\cdot\mathbf{E}_2. \tag{3.97}$$

如果 $\nabla\cdot$ E = 0，则向量场 E 称为无散。

方程给出的结果。对于微分体积 $\Delta\upsilon$ 的方程（3.93）可以扩展为将任意体积 υ 上的 $\nabla\cdot$ E 的体积积分与 E 通过包围 υ 的闭合表面 S 的通量联系起来。那是，

$$\int_\upsilon \nabla\cdot\mathbf{E}\,d\upsilon = \oint_S \mathbf{E}\cdot d\mathbf{s}. \tag{3.98}$$

(divergence theorem)

这种关系被称为散度定理，广泛应用于电磁学中。

例 3-13：计算散度

确定以下每个向量场的散度，然后在指定点对其进行评估:

(a) $\mathbf{E} = \hat{\mathbf{x}}3x^2 + \hat{\mathbf{y}}2z + \hat{\mathbf{z}}x^2z$ at $(2,-2,0)$;
(b) $\mathbf{E} = \hat{\mathbf{R}}(a^3\cos\theta/R^2) - \hat{\boldsymbol{\theta}}(a^3\sin\theta/R^2)$ at $(a/2, 0, \pi)$.

解： (一)

$$\begin{aligned}\nabla\cdot\mathbf{E} &= \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}\\ &= \frac{\partial}{\partial x}(3x^2) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}(x^2z)\\ &= 6x + 0 + x^2 = x^2 + 6x.\end{aligned}$$

At $(2,-2,0)$, $\nabla\cdot\mathbf{E}\big|_{(2,-2,0)} = 16.$